Lab 11
Case Study

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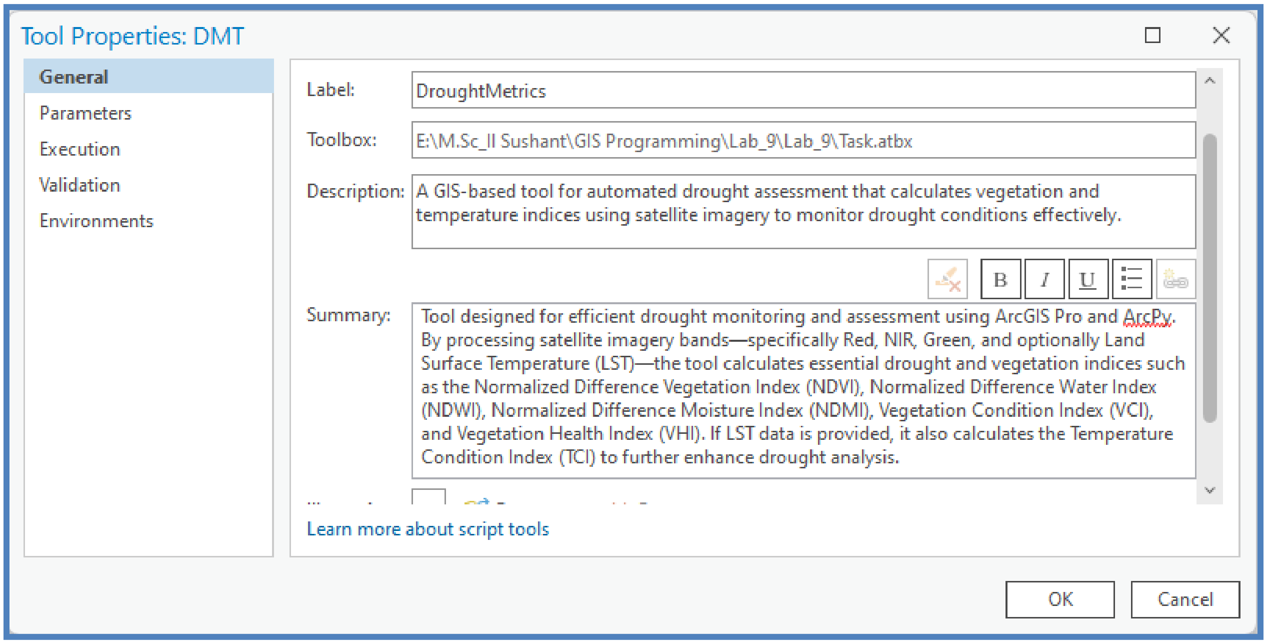
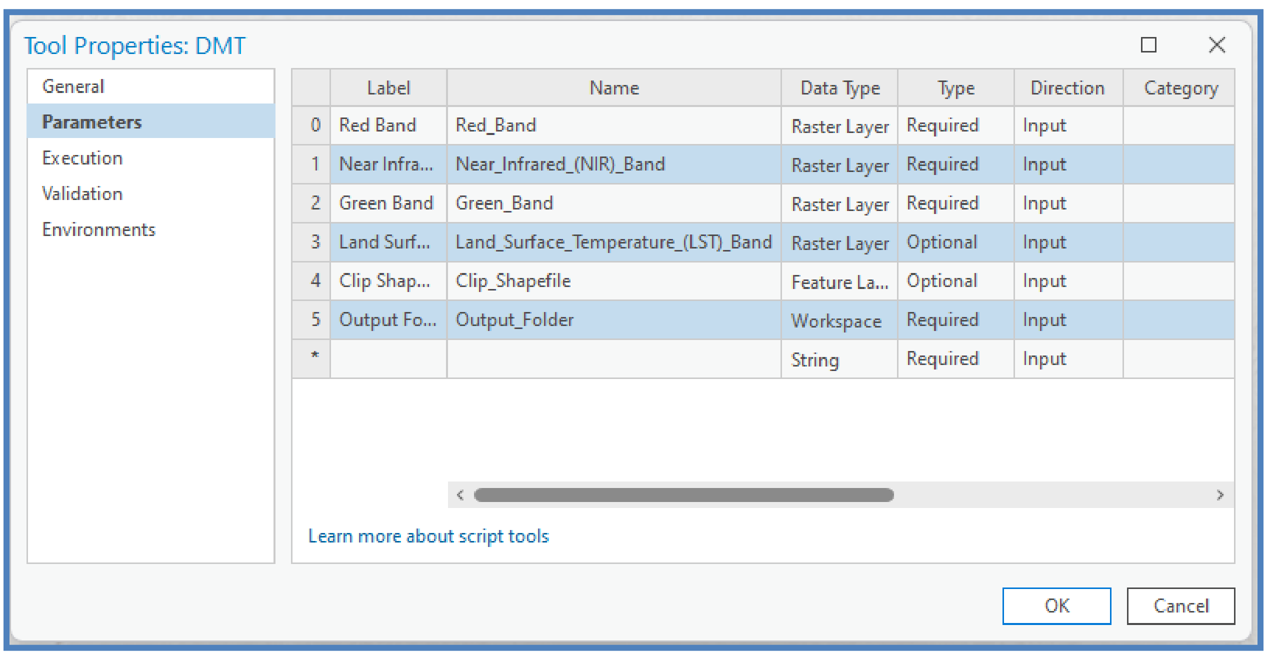
To develop a GIS-based tool that can monitor drought conditions using vegetation and temperature indices.



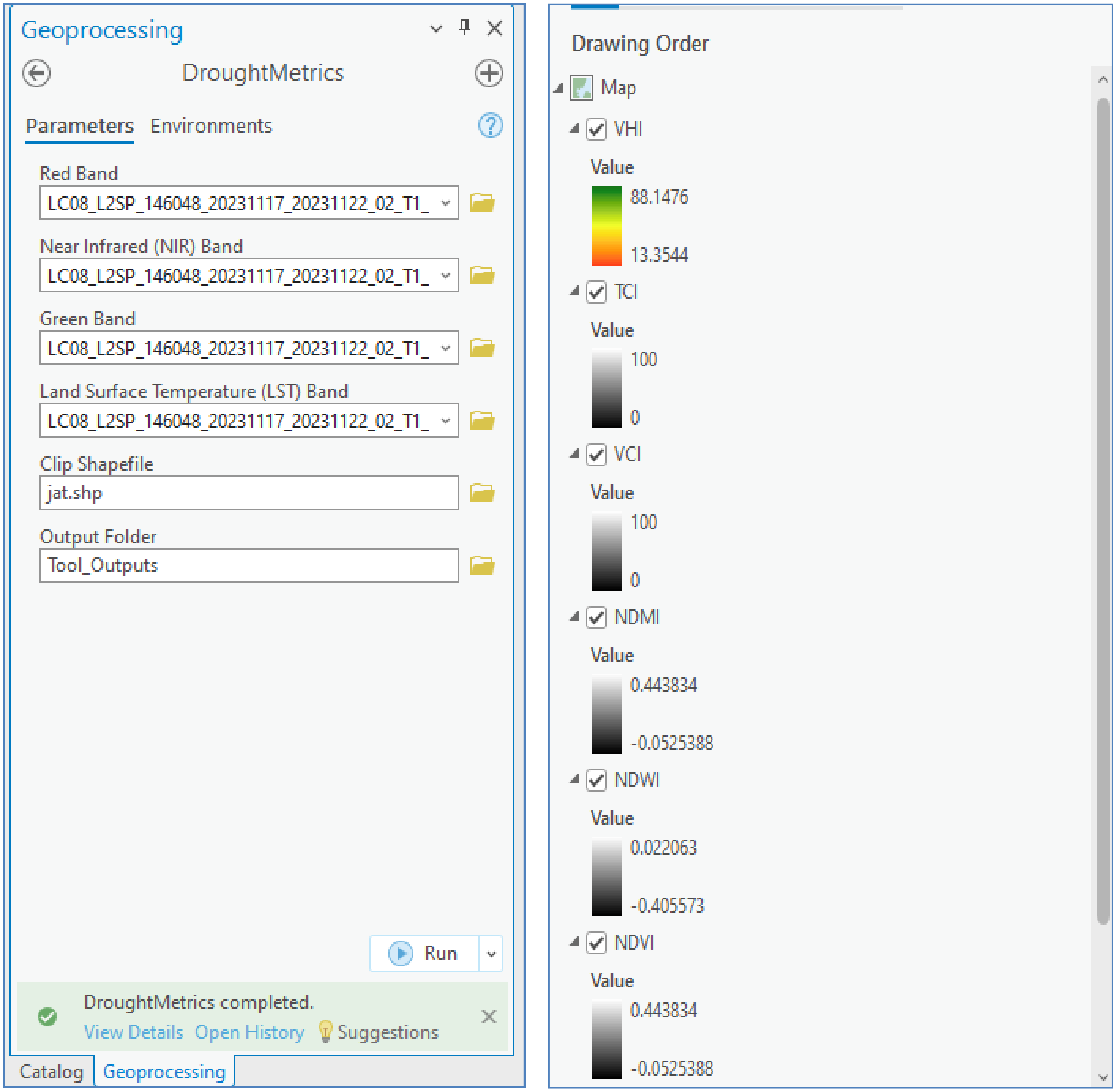


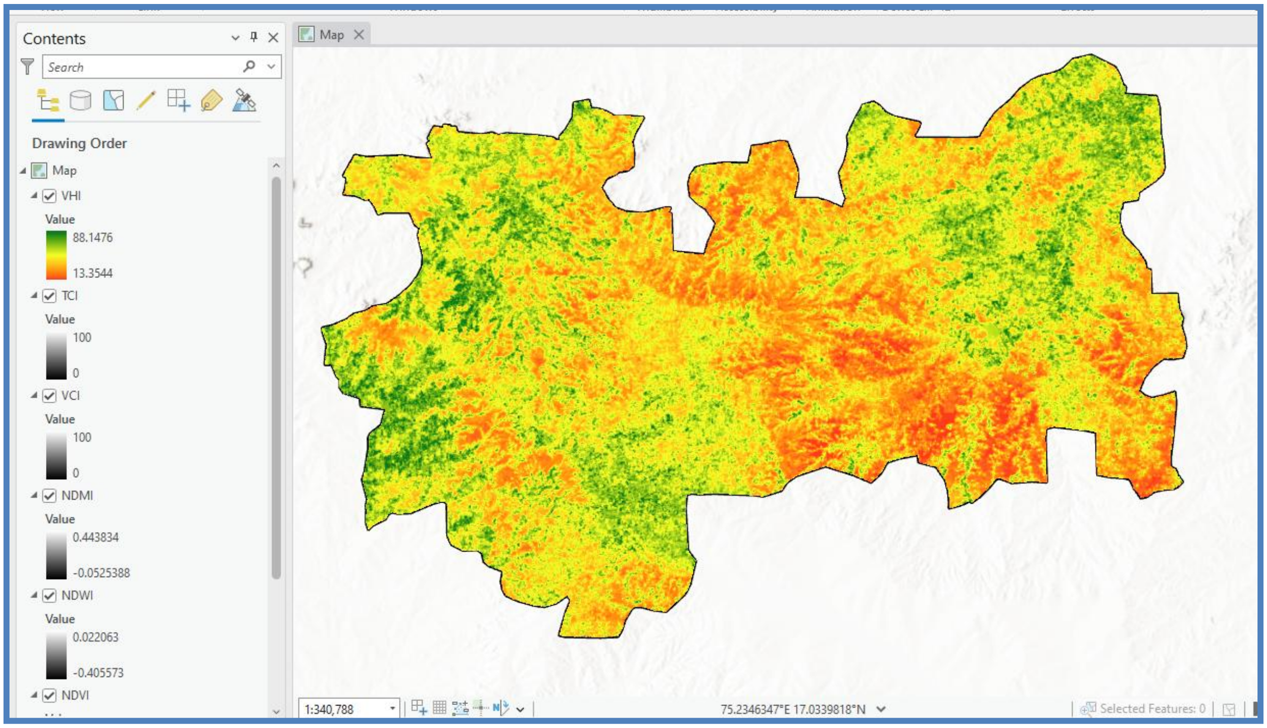
Tool Properties: DMT

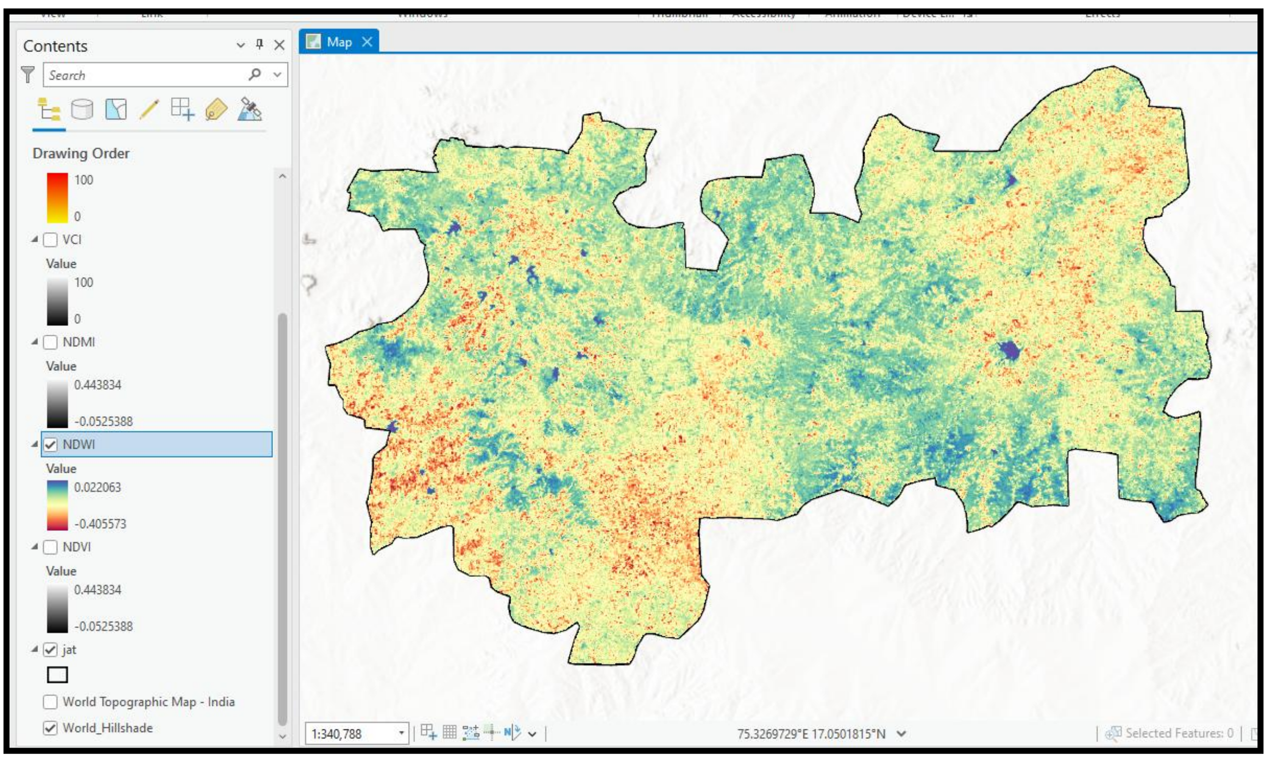
- General
- Parameters
- **Execution**
- Validation
- Environments

Script File ... embedded ...

```
import arcpy
from arcpy.sa import *

# Set input parameters
red_band = arcpy.GetParameterAsText(0)       # Red band (required)
nir_band = arcpy.GetParameterAsText(1)       # NIR band (required)
green_band = arcpy.GetParameterAsText(2)     # Green band (required)
lst_band = arcpy.GetParameterAsText(3)       # LST band (optional)
clip_shapefile = arcpy.GetParameterAsText(4) # Clip area shapefile (optional)
output_folder = arcpy.GetParameterAsText(5)  # Output folder

# Set output paths
ndvi_output = output_folder + "/NDVI.tif"
ndwi_output = output_folder + "/NDWI.tif"
ndmi_output = output_folder + "/NDMI.tif"
vci_output = output_folder + "/VCI.tif"
vhi_output = output_folder + "/VHI.tif"
tci_output = output_folder + "/TCI.tif"

# Check out the Spatial Analyst extension
arcpy.CheckOutExtension("Spatial")

# Clip raster inputs if a shapefile is provided
def clip_raster(raster):
    if clip_shapefile:
        return ExtractByMask(Raster(raster), clip_shapefile)
    return Raster(raster)

# Clip the bands
red_band = clip_raster(red_band)
nir_band = clip_raster(nir_band)
green_band = clip_raster(green_band)
lst_band = clip_raster(lst_band) if lst_band else None

# Calculate NDVI
ndvi = (nir_band - red_band) / (nir_band + red_band)
ndvi.save(ndvi_output)

# Calculate NDWI
ndwi = (green_band - nir_band) / (green_band + nir_band)
ndwi.save(ndwi_output)

# Calculate NDMI (assuming SWIR is red here - confirm with your data)
```


Geoprocessing
DroughtMetrics
Parameters
Environments
Red Band
LC08_L2SP_146048_20231117_20231122_02_T1_
Near Infrared (NIR) Band
LC08_L2SP_146048_20231117_20231122_02_T1_
Green Band
LC08_L2SP_146048_20231117_20231122_02_T1_
Land Surface Temperature (LST) Band
LC08_L2SP_146048_20231117_20231122_02_T1_
Clip Shapefile
jat.shp
Output Folder
Tool_Outputs
Run
DroughtMetrics completed.
View Details
Open History
Suggestions
Catalog
Geoprocessing
Drawing Order
Map
VHI
Value
88.1476
13.3544
TCI
Value
100
0
VCI
Value
100
0
NDMI
Value
0.443834
-0.0525388
NDWI
Value
0.022063
-0.405573
NDVI
Value
0.443834
-0.0525388


Contents
Search
Map
Drawing Order
Map
VHI
Value
88.1476
13.3544
TCI
Value
100
0
VCI
Value
100
0
NDMI
Value
0.443834
-0.0525388
NDWI
Value
0.022063
-0.405573
NDVI
1:340,788
75.2346347°E 17.0339818°N
Selected Features: 0


Contents
Search
Map
Drawing Order
100
0
VCI
Value
100
0
NDMI
Value
0.443834
-0.0525388
NDWI
Value
0.022063
-0.405573
NDVI
Value
0.443834
-0.0525388
jat
World Topographic Map - India
World_Hillshade
1:340,788
75.3269729°E 17.0501815°N
Selected Features: 0